Engineering Mathematics - I · Unit_05 · MATLAB

DRAFT — UE26MA141A · Unit_05 · MATLAB · Video

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

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M1

Video

1 Video

Study focus: Video. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000098 · EV0000099

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session we are going to learn about higher order differential equations in MATLAB.

Key details

- Here our first question to solve the higher order differential equations that is y double dash plus phi y which is equals to zero.
- Things of y of x which is nothing but we are.
- Defining the variables I mean symbolic variables next ODE which is equals to diff of y comma x comma 2 plus phi star fi double is equals to zero.
- Here O D is nothing but ordinary differential equations and d if f is nothing but differentiation and d if f of y comma x comma two is nothing but second derivy q plus phi into y which is double is equal to zero.
- Where double is equals to which means defining an equation symbolically that is left hand side which is equals to right hand side.
- Next to for the solution we are taking y solve of x which is equals to d solve of ODE.
- Where d solve of.
- Where D solve which is equal to differential equation solver of ODE function.
- First we are defining the symbols that is y of x.
- Next, we are defining the question that ODE which is equals to differentiation of y, comma x, comma 2.
- 5 star y which is double is equals to 0.
- Semicolon next y solution of x which is equals to D sol of Ode.
- Moving on to the similar type of the question that is y double dash plus seven y which is equals to zero and we'll get the solution that is y solution of x which is equals to c1 cos root seven x minus c2 sin root seven x.

- This is of similar portion instead of 7 I have taken 18y which is equal to 0.
- Next moving to the higher order higher order differential equation.
- That is 12y double dash plus 10y dash minus 8y which is equals to 0.
- Here again we are defining the variables that is of since of y of x and od which is equal to 12 star.
- The sense 12 into differentiation this is of second derivative plus 10 into first derivative minus 8y which is double is equals to 0 and y solve of x is equals to d solve of od.
- Next moving on to the matlab I'm taking a section break.
- This is of defining the symbols Simps of Y of X.
- Next, defining the question.
- Defining the question as ODE which is equal to 12 star second derivative d i f of y comma x comma to plus 10 star differentiation of y, comma x comma 1, which is the first derivative minus 8 star y.
- And double is equal to 0.
- I am closing with semicolon.
- Next moving on to the Y solution of X, which is equals to D solve of ODE.
- x by 2 plus even e power minus 4x by 3.
- Next moving on to the next question of similar question that is 8y double dash plus 2y dash minus y is equal to 0 as of same question.
- Next this is of order 4.
- Hereof also seen we are defining the symbols that is of seams of y of x and od is equals to differentiation of fourth order of.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 05:00–09:59

Fourth the different derivative differentiation of y, comma four plus eight into second.

Key details

- Second derivative plus 16y which is equal to 0 and y solve of x is equals to d solve of ODE.
- Here of order 3 of similar question.
- Next here I have taken d cube minus 3 d square plus 4.
- into x which is equal to 0.
- Hereof also same we are defining the symbolic variables that is y of x and o d e which is equals to third order derivative minus 3 into second derivative.
- Plus 4y which is equals to double is equals to 0 and y solution which is equals to desolve of y.
- Here I am defining the symbols.
- This is of y of x.
- Next defining the ODE function ODE which is equal to DIFF of differentiation of Y, X, comma, 3.
- Minus 3 star d if of differentiation of y, x, 2.
- Plus 4 star y double is equals to 0.
- I am closing with semicolon and y solve.

- Y solve of x which is equals to D solve of Ode.
- I'm getting output as C1 E power 2X plus C3 E power minus X plus C2 X E power 2X.
- Next, moving on to the next question.
- Here thereof initial conditions.
- First here defining as the symbols of variables.
- Sims of y of x and we are defining the question as ODE which is equals to second derivative.
- Minus 5 into first derivative plus 6y which is double is equals to 0 and dy which is equal to first derivative we are defining and d2y which is equal to second derivative we are defining.
- And C O N D 1, which is of condition 1, which is equal to y of 0, double is equals to minus 1, and condition 2, which is equals to dy of 0, double is equals to 0.
- And y solution instead.
- We are taking all the three conditions that is of D solve of ODE, condition one, comma condition two.
- Here Dispov Y solve, which is nothing but we are displaying the.
- and EZ of EZ plot of Y sort which is nothing but easy to plot the graph.
- So first we are defining the scenes since of x.
- Next define the question as ODE which is equal to DIFF of y comma x comma two minus 5 into PIF of y, comma x.
- This is a this is our first derivative plus 6 star y which is double is equals to 0.
- I am again closing with semicolon and.
- First derivative which is equal to dif of y, comma x and d2 y which is equal to second derivative.
- Dif of y, comma x, comma two and I am defining the conditions.
- It is of condition one which is equal to dy which is equals to or y of 0.
- Y of 0 which is equals to minus 1.

Example or application from the lecture

- Let us see in MATLAB code.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 10:00–14:59

Dou is equals to minus one.

Key details

- So second condition two which is equal to D2y.
- Dy of zero, which double is equals to zero.
- I'm closing with semicolon.
- Next Y solve Y solution of X.
- which is equal to D solve of ODE, comma, condition one, comma, condition two.
- I am taking all the three conditions.
- and closing with semicolon and I am displaying the solution display of y solution so I am getting the output that is 2 e power 3x minus 3 e power 2x and for the plot I am using.

- E Z plot that is of easy to plot the Y solution.
- Next moving on to the similar questions.
- And the question is y triple dash minus 6y double dash plus 1 y dash minus 6y is equals to 0.
- For this question also, we are going to plot the graph.
- You are getting this type of graph.
- Next moving on to the single question.
- That is of conditions with y of 1 is equals to 0 and y dash of 1 is equals to minus 1.
- Next question, next conditions you are defining.
- Next solution, you are displaying the solution or you are simplify the solution.
- Next moving on to the a z plot of solution.
- Here I have taken one step extra that is why solution is equals to simplify of y solve.
- This is nothing but we are getting the output and we are solving again the same solution and we are displaying the solution.
- We are plotting for the graph for the same question.
- This type of answer we are going to get that is minus e power 2 minus 2x of x minus 1.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Video transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Video transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Video transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.
3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.
4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
11. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...
SRC00023	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...
SRC00025	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare...

C Revision and retrieval practice